

EdLearn Platform Engineering Report

Comprehensive Architectural Audit, Upgraded Systems & Remaining Gaps

Date: August 20, 2026 | Status: Production-Grade (84/84 Tests Passed, 0 Typecheck Errors)

1. Executive Summary

EdLearn is an enterprise-grade educational platform built with Next.js 16, React 19, TypeScript, Express, Prisma 7.8, PostgreSQL, Redis, and multi-provider AI pedagogies. Following our architectural audit, 5 parallel engineering streams successfully completed high-value feature tracks without merge conflicts or regressions.

2. Fully Upgraded Platform Subsystems

Member 1: Content & Discovery

- Universal Bookmarks: Full-stack persistence across books, media videos, podcasts, technology news, and study topics with dedicated /bookmarks collection hub.
- Universal Global Search (Cmd+K): Fast, debounced modal querying all 5 database content types simultaneously with keyboard navigation.

Member 2: Learning Progress & Gamification

- Streak Calculation Engine: Timezone-aware active and longest streak tracking with weekly target completion bars.
- 365-Day Activity Heatmap: GitHub-style contribution calendar displaying daily study consistency and lesson completions on the dashboard.

Member 3: Assessment & Engagement

- Interactive Quiz Runner: Multi-question diagnostic evaluation embedded into LivingDocument with instant scoring, feedback, and celebration animations.
- Personal Notes Scratchpad: Markdown scratchpad with debounced real-time auto-saving and formatting tools.

Member 4: User, Community & Administration

- Role-Based Access Control (RBAC): Role hierarchy (USER, ADMIN, MODERATOR) protecting management routes.
- Admin Oversight Portal: Comprehensive platform telemetry dashboard at /admin with user role promotion and stats.
- In-App Notification Center: Real-time achievement, streak, and milestone notification dropdown.

Member 5: Platform, Security & Engineering

- AI Secondary Failover: Automatic circuit breaker switching between Groq (Llama-3.3-70b) and Google Gemini upon rate limits or 5xx outages.
- Credit Guards: Pre-flight credit verification protecting AI endpoints.

3. Remaining Gaps & Next-Phase Opportunities

While the core individual study experience, assessment runner, content hubs, and administration are complete, the following product gaps represent high-impact areas for future phases:

- Gap 1: Real-Time Collaborative Study Rooms: Current platform supports single-user study. Opportunity: Add WebRTC/Socket.io live study rooms with synchronized audio playback and shared whiteboard notes.
- Gap 2: PWA Offline Sync & Local Cache: Current platform requires active network. Opportunity: Service worker caching of study notes and background queueing for offline quiz submissions.
- Gap 3: Spaced Repetition Flashcards (SM-2): Current quiz scores are saved, but mistakes are not scheduled for review. Opportunity: Automatically convert missed quiz questions into SuperMemo-2 spaced repetition review decks.
- Gap 4: Social Discussion Threads per Topic: Current study notes are private. Opportunity: Embed threaded community discussions and peer Q&A at the bottom of each study topic.
- Gap 5: Automated Email Digests & Weekly Summaries: Current notifications are in-app only. Opportunity: Scheduled weekly digest emails summarizing streak records, completed roadmaps, and recommended topics.
- Gap 6: Verifiable Course Certificates: Current course completions award badges. Opportunity: PDF certificate generator with unique verification hashes and QR codes for LinkedIn sharing.

4. Verification & Quality Metrics

- Vitest Test Suite: 84/84 tests passing across all 6 test suites (auth, content, gdpr, generate, roadmap, platform_features).
- TypeScript Compliance: 0 errors across both frontend (npx tsc --noEmit) and backend (npm run typecheck).
- Database Consistency: Prisma 7.8 schema verified with foreign key constraints, cascade rules, and compound indexes.